

AI HR RECRUITMENT ASSISTANT

An Agentic AI based solution for Intelligent, Automated Talent Acquisition

IBM Internship Project — Agentic AI

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1. Project Overview

1.1 Project Title:

AI HR Recruitment Assistant — An Agentic AI System for Automated, Intelligent, and Candidate-Centric Hiring.

1.2 Problem Statement:

Traditional recruitment processes depend heavily on manual resume screening and repeated Level-1 (L1) coordination effort by HR staff for routine tasks such as shortlisting candidates, scheduling interviews, answering applicant FAQs, and following up on offer status. This manual approach leads to long time-to-hire, inconsistent screening quality, poor candidate experience, and a high operational load on recruiters. Talent teams often take days to respond to even simple candidate queries, and struggle to scale hiring during peak recruitment drives or when handling high applicant volumes.

1.3 Brief Description of the Project:

The AI HR Recruitment Assistant is an Agentic AI system built to autonomously handle the end-to-end lifecycle of routine recruitment tasks. Instead of a single monolithic chatbot, the solution uses a team of cooperating AI agents — each responsible for a specific task such as parsing resumes, matching candidates to job requirements, scheduling interviews, answering candidate queries, and escalating complex or sensitive decisions to human recruiters. A central orchestrator agent plans the workflow, delegates tasks to the right specialised agent, and adapts its approach based on intermediate results, allowing the system to handle common recruitment activities autonomously while keeping humans in control of final hiring decisions.

2. Objectives & Proposed Solution

2.1 Project Objectives:

G Automate resume screening and candidate-job matching to reduce manual effort on the HR team.

G Reduce Time-to-Hire (TTH) for common, high-volume recruitment roles.

G Provide instant, 24x7 self-service support to candidates through a conversational agent.

G Reduce the repetitive workload on recruiters so they can focus on final-stage decisions and candidate relationships.

G Automatically handle routine tasks such as interview scheduling, status updates, and FAQ resolution.

G Give HR operations teams a real-time dashboard with pipeline and hiring-funnel analytics.

The system follows a multi-agent architecture orchestrated by a central Planner Agent, built on a large language model (LLM) with tool-calling capability. When a candidate applies through a careers portal, email, or job board, the Planner Agent breaks the request into sub-tasks and routes them to specialised agents:

G Intake Agent: Captures the application from the careers portal, email, or job board and normalises it into a structured candidate profile.

G Resume Parsing Agent: Uses NLP to extract skills, experience, education, and other structured fields from resumes in varied formats.

G Matching & Ranking Agent: Performs Retrieval-Augmented Generation (RAG) over job descriptions and past hiring data to score and rank candidates against open roles.

G Screening Agent: Runs automated checks such as eligibility criteria, keyword and skill-fit validation, and duplicate-application detection to shortlist candidates.

G Scheduling Agent: Coordinates interview scheduling automatically, such as sending calendar invites, confirming slots, and sending reminders.

G Candidate Engagement Agent: Answers candidate FAQs (application status, role details, company policies) and sends timely status updates, avoiding repeated queries to recruiters.

G Escalation Agent: Routes ambiguous, sensitive, or final-round cases to a human recruiter with the full candidate context attached.

G Planner / Orchestrator Agent: Coordinates the above agents, monitors outcomes, retries failed steps, and keeps both candidates and recruiters informed throughout.

Human-in-the-loop review is retained for final hiring decisions and escalated cases, so the AI accelerates and standardises the early stages of recruitment while critical decisions remain with the HR team.

2.3 Key Features:

G NLP-based resume parsing and automatic candidate-role matching with ranking scores.

G RAG-based knowledge base search for instant, accurate answers to candidate and job queries.

G Automated scheduling for common tasks — interview coordination, reminders, and status updates.

G Seamless escalation to human recruiters with full conversational and candidate-profile context handoff.

G 24x7 multi-channel support via chat, email, and platforms such as Slack or Microsoft Teams.

G Real-time HR operations dashboard with hiring-funnel tracking and recruitment analytics.

3. Implementation & Results

3.1 Technologies / Tools Used:

G Programming Language: Python

G Agent Orchestration: LangChain / Agentic workflow framework

G LLM & AI Services: IBM watsonx.ai (foundation models), NLP with spaCy / transformer models

G Knowledge Retrieval: Vector database (e.g. Chroma / Milvus) for RAG over job descriptions and HR policy documents

G Backend / API: Flask / FastAPI

G Frontend / Dashboard: Streamlit

G Database: SQLite / PostgreSQL

G Integrations: Applicant Tracking System API (e.g. Workday / Greenhouse), Calendar API (Google Calendar / Outlook), Slack / Teams API, SMTP

G Version Control & Deployment: Git, Docker

3.2 Working Process:

G **Step 1 — Intake:** The candidate applies via the careers portal, email, or a job board.

G **Step 2 — Resume Parsing:** The Resume Parsing Agent extracts structured candidate data from the application.

G **Step 3 — Matching & Ranking:** The Matching & Ranking Agent scores the candidate against relevant open roles.

G **Step 4 — Screening:** The Screening Agent runs automated eligibility and skill-fit checks to shortlist candidates.

G **Step 5 — Scheduling:** The Scheduling Agent automatically coordinates interviews if the candidate is shortlisted.

G **Step 6 — Escalation:** If the case is ambiguous or reaches a final round, the Escalation Agent hands it off to a human recruiter with full context.

G **Step 7 — Reporting:** The HR dashboard updates pipeline status, funnel metrics, and hiring analytics in real time.

3.3 Screenshots / Output:

[Insert screenshots here demonstrating the working chatbot interface, a sample candidate-matching run, and the HR operations dashboard, as proof of execution.]

Sample processed-application log generated by the Matching & Ranking Agent:

3.4 Results Achieved:

The prototype was evaluated on a sample set of candidate applications against the existing manual process, with the following observed improvements:

Metric	Before (Manual Process)	After (AI Agent)	Improvement
Avg.screeningtime/candidate	15-20 minutes	~1.5 minutes	~90% faster
Applicationsprocessed/day/recruiter	40-60	500+	~9x throughput
Shortlist accuracy (role fit)	Baseline	85%	High consistency
Viewno-show/schedulingerrorrate	18%	~5%	Reduced coordination load
Candidatequeryresponsetime	Hours to days (email/queue)	Instant (chatbot)	24x7 availability

These results indicate that the agentic pipeline significantly reduces the manual workload on HR staff while maintaining reliable, consistent screening, allowing recruiters to focus their time on final-round evaluation and candidate relationships rather than repetitive first-stage screening.

Application ID	Role Applied	Screening Status	Handled By	Time Taken
APP-2201	Software Engineer	Shortlisted	Matching & Ranking Agent	50 sec
APP-2202	Data Analyst	Shortlisted	Matching & Ranking Agent	1 min 10 sec
APP-2203	Senior Manager	Escalated	Human Recruiter	22 min
APP-2204	HR Executive	Rejected (Eligibility)	Screening Agent	40 sec

4. Conclusion & Future Scope

4.1 Project Conclusion:

The AI HR Recruitment Assistant demonstrates how Agentic AI — a system of cooperating, tool-using AI agents rather than a single static chatbot — can meaningfully automate the early, repetitive stages of recruitment. By dividing responsibilities among specialised agents coordinated by an orchestrator, the system achieves faster, more consistent, and more scalable candidate processing while keeping final hiring decisions in human hands.

4.2 Challenges Faced:

- G Handling inconsistent resume formats and incomplete candidate information during parsing.
- G Integrating reliably with legacy Applicant Tracking Systems (ATS) and calendar platforms.
- G Ensuring fairness and avoiding bias in automated candidate screening and ranking.
- G Avoiding incorrect auto-rejections and ensuring safe fallback to human recruiter review.

4.3 Future Enhancements:

- G Add a voice-based recruitment assistant for phone-screening interactions.
- G Introduce predictive analytics to forecast candidate drop-off and offer-acceptance likelihood.
- G Extend integration to additional ATS platforms and enterprise HRIS systems.
- G Incorporate continuous learning from hiring outcomes and recruiter feedback to improve matching accuracy over time.
- G Add multilingual support for global or distributed hiring teams.

4.4 References:

1. IBM watsonx.ai Documentation — <https://www.ibm.com/watsonx>
2. LangChain Documentation — <https://python.langchain.com>
3. spaCy NLP Library — <https://spacy.io>
4. Applicant Tracking System (ATS) API Documentation.
5. Research literature on Agentic AI architectures and multi-agent LLM systems.
6. Internal IBM Internship — Agentic AI project guidelines.